Beyond Spatial Priors: Spectral-Aware Hyperspectral Anomaly Detection Network via Heterogeneous–Homogeneous Super-Band and Quaternionic Sparse Representation

Xianyue Wang, *Graduate Student Member, IEEE*, Yuli Sun, Tao Tang, and Lin Lei

***Abstract*—When incorporating spatial priors (such as sparsity and low-occurrence frequency) to identify anomalies, existing hyperspectral anomaly detection (HAD) methods typically operate on the original high-dimensional spectral bands or features obtained through dimensionality-reduction techniques. However, they often neglect the explicit modeling and exploitation of the internal discriminative structure inherent in bands, consequently diluting the discriminative spectral information crucial for effective anomaly-background separation. This limitation becomes particularly pronounced when processing hyperspectral data, where the rich spectral heterogeneity could provide valuable discriminative cues for HAD. To address this issue, we propose a super-band quaternion-guided sparse network (SuperB-QSNet), which explicitly encodes the spectral discriminant structure while retaining the advantage of spatial sparse priorities. First, a heterogeneous–homogeneous super-band compression (HH-SBC) organizes spectral bands into meaningful heterogroups. It naturally reduces redundancy and retains the representative spectral information. Second, the quaternionic sparse recovery (QSR) model encodes both spatial structures and intragroup spectral correlations via hypercomplex algebra, ensuring robust anomaly preservation and enhanced detection of spectral variations. Third, we design a difference-guided quaternionic convolution detector (DG-QCD) to maintain spectral coherence via group processing and enhance discriminative capability via intergroup difference operation. Extensive experiments show that SuperB-QSNet enhances anomaly detection accuracy while ensuring computational efficiency across real-world hyperspectral datasets.**

***Index Terms*—Hyperspectral anomaly detection (HAD), quaternion representation, spatial and spectral priors, superpixel segmentation.**

NOMENCLATURE

$\mathbf{H} \in \mathbb{R}^{r\times c\times b}$	Hyperspectral image (HSI).
$\boldsymbol{H} \in \mathbb{R}^{s\times b}$	Unfolding a 2-D matrix of HSI.
$\boldsymbol{h}_i \in \mathbb{R}^{s\times 1}$	ith spectral vector in $\boldsymbol{H}$.
$\boldsymbol{p}^j \in \mathbb{R}^{b\times 1}$	Spectral observations of the jth pixel.
$\boldsymbol{M}_i^{(m)} \in \mathbb{R}^{b\times\sigma}$	Spectral matrix for cross-segmentation.
$\boldsymbol{t}_i^{(m)} \in \mathbb{R}^{1\times n}$	Index sets of mutation points.
$\boldsymbol{t}^* \in \mathbb{R}^{1\times n}$	Final index set of mutation points.
$\boldsymbol{B}_j \in \mathbb{R}^{s\times 4}$	Selected band set of the jth super-band.
$\mathbf{B}_j \in \mathbb{R}^{r\times c\times 4}$	Folding $\boldsymbol{B}_j$ to conduct spatial recovery.
$\mathcal{B}^j \in \mathbb{Q}^{r\times c}$	Quaternization matrix of $\mathbf{B}_j$.
$\mathcal{A}^j \in \mathbb{Q}^{r\times c}$	Sparse anomaly part of $\mathcal{B}^j$.
(μ, ν)	Spatial position of any pixel.
$\mathbf{A}^j \in \mathbb{R}^{r\times c\times 4}$	Sparse anomaly part in the real-number field.
$\mathbf{A} \in \mathbb{R}^{r\times c\times 4n}$	Global sparse component.
$G_\delta(\cdot)$	Convolution function.
$\mathbf{K} \in \mathbb{R}^{1\times 1\times r\times c}$	Anomaly score map.

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I. INTRODUCTION

IN REMOTE sensing images, an anomaly typically refers to a small spatial region that exhibits distinct characteristics compared to its surrounding background [1]. These anomalies can arise from natural phenomena (e.g., mineral leakage and vegetation diseases) or artificial targets (e.g., camouflage facilities and building ruins) and can be characterized by two key features: 1) sparse spatial distribution, with occurrence frequencies much lower than background pixels, and 2) subtle spectral differences, making them challenging to detect [2], [3], [4]. Anomaly detection aims to isolate these rare yet distinctive targets from complex and heterogeneous backgrounds without prior category labels.

Hyperspectral images (HSIs), with their continuous narrow spectral bands, provide significant advantages for anomaly detection. HSIs capture detailed reflection characteristics of various substances across spectral bands, enabling precise identification of subtle spectral variations [4], [5]. They enhance the detection of subtle anomalies by leveraging spatial and spectral information, even those with minimal spectral deviations or subpixel coverage [6], [7]. This multidimensional analysis combines local spatial distribution, global contextual features, and fine-grained spectral characteristics, significantly improving detection accuracy. Therefore, hyperspectral anomaly detection (HAD) has become indispensable in numerous fields, such as ecological monitoring [8], mineral exploration [9], and military reconnaissance [10]. It should be noted that HAD extends the capabilities beyond Earth, facilitating interstellar exploration by identifying spectral anomalies in planetary surfaces [9]. This broad applicability underscores HAD's significance as a powerful terrestrial and extraterrestrial analysis tool.

HAD algorithms can be systematically categorized into three primary groups based on their underlying mechanisms [4]. First, statistical-based methods primarily depend on statistical models of spectral variance and covariance structures, operating under the premise that anomalies exhibit significant deviations from the statistical properties of the background. A higher order adaptive cosine estimator-based detector overcomes the limitations of the assumption of a normal multivariate distribution of background in the Reed–Xiaoli (RX) [11] detector. A fractional Fourier entropy-based model (FrFE-RX) is also developed, incorporating the fractional Fourier transform with RX [12]. Nevertheless, the detection performance of these methods frequently degrades significantly in complex environments with diverse and nonuniform background characteristics.

Second, representation-based methods [13], capitalizing on the sparse spatial distribution of anomaly targets, typically employ the principle that background pixels can be effectively represented by a learned low-rank dictionary, whereas anomalies cannot. Qin et al. [14] developed a generalized nonconvex low-rank tensor representation (GNLTR) for HAD, which utilizes double low-rank recovery processing to obtain the background and corresponding dictionary. Additionally, SuperRPCA employs a superpixel-wise collaborative representation detector to reconstruct the background and a robust principal component analysis (RPCA) to further separate the background from anomaly pixels [15].

Finally, deep learning models based on low-occurrence frequency typically analyze background reconstruction errors, where more significant errors indicate a higher likelihood of anomaly pixels. Traditional approaches focus on learning the background distribution specific to individual datasets, which limits their generalizability across different scenarios [16], [17]. In contrast, Li et al. [4], [18], [19] proposed a novel cross-modality anomaly detection framework that leverages a consistent deviation metric for score ranking. This approach shifts the paradigm from learning specific anomaly score values to learning the relative ranking of scores, akin to the transformation from parametric to nonparametric statistics [20]. While this innovation enhances the intrinsic connection between multimodal data, its underlying principle remains rooted in the fundamental distinction between background and anomaly characteristics and the low-occurrence frequency of anomalies across multiple remote sensing images. Besides, compared to statistical-based methods, representation- and deep learning-based approaches demonstrate excessive dependence on the purity of background assumptions in their mask configurations or dictionary constructions [21].

Although these algorithms have effectively leveraged the fundamental spatial prior knowledge (sparse distribution and low-occurrence frequency) of anomaly detection, they frequently fail to fully exploit the intrinsic spectral structure of HSIs that complements these spatial assumptions. Specifically, most methods perform global processing operations that treat the entire high-dimensional spectral band set uniformly (e.g., computing global covariance matrices, applying full-band low-rank decomposition, or processing all bands as independent channels), thereby overlooking the physically driven organization of spectra into homogeneous groups separated by heterogeneous boundaries [12], [14], [18]. Spectral homogeneity refers to the commonality among correlated spectral bands, enabling natural dimensionality reduction by identifying consistent spectral patterns [22], [23], [24]. Meanwhile, spectral heterogeneity, characterized by the discriminative variance among distinct band groups, provides crucial information for material identification [25], [26]. When these complementary structural properties are not explicitly modeled, discriminative spectral features that could aid anomaly-background separation become diluted in global processing [27]. This limitation becomes particularly critical when processing hyperspectral data containing rich homogeneous–heterogeneous spectral channels, where the structural information could provide valuable discriminative cues beyond spatial sparsity alone [18], [19].

To overcome the aforementioned issues, we propose the super-band quaternion-guided sparse network (SuperB-QSNet) composed of the following three modules. First, to overcome the overlook of spectral priors, we design a heterogeneous–homogeneous super-band compression (HH-SBC), which groups bands based on their intrinsic characteristics rather than arbitrary clustering metrics. The method first applies a cross-segmentation algorithm that extends superpixel concepts to the spectral domain. A parallel quantile-based band selection obtains four representative bands per group, reducing redundancy and enhancing discriminative features for anomaly detection. Second, focusing on intragroup relationships of homogeneous spectral bands rather than employing global processing of all bands (which may disrupt the original spectral structures and introduce unnecessary interference), our quaternionic sparse recovery (QSR) model encodes each super-band as a quaternion. It eliminates the need for explicit background modeling and delivers a refined sparse input to the subsequent networks, thereby improving sensitivity to subtle spectral deviations. Finally, to leverage the spectral–spatial features from the preceding components, we design a difference-guided quaternionic convolution detector (DG-QCD) that processes band groups holistically. It further amplifies discriminative features by computing differences between adjacent heterogeneous groups, enhancing anomaly-background separation. The framework is designed as a streamlined deep detection pipeline that operates without supervision, though not strictly end-to-end due to the nondifferentiable band grouping step.

The main contributions of this article can be summarized as follows.

1) We propose a framework unifying spectral compression, sparse recovery, and detection modules to develop a complete data-mechanism co-driven paradigm for HAD.
2) Our framework maintains spectral structural integrity throughout all processing stages by designing operations that jointly preserve spatial arrangements and spectral grouping in an inseparable manner.

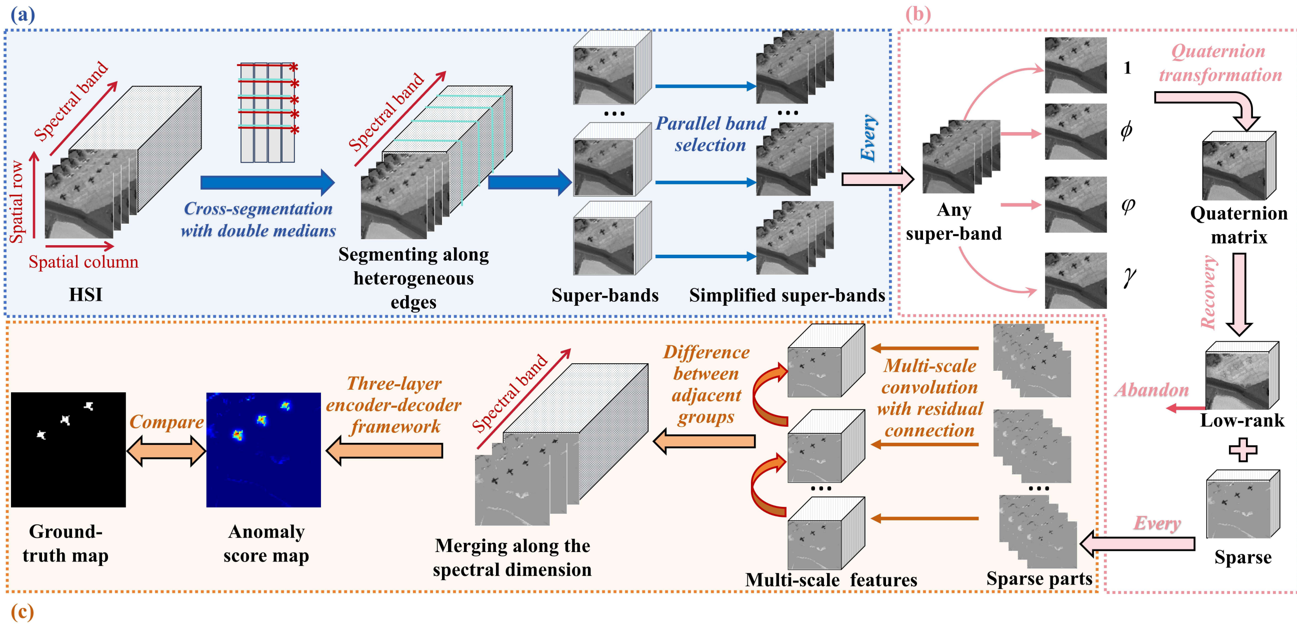

Fig. 1. Flowchart of the proposed SuperB-QSNet: A HAD framework integrating (a) HH-SBC yielding optimized super-bands through heterogeneous edge segmentation and parallel band selection, (b) QSR producing low-rank background and sparse components via quaternion transformation, and (c) DG-QCD generating the final anomaly score map by exploiting intergroup spectral differences and multiscale convolution features.

3) We extend spatial algorithms to the spectral domain, introducing heterogeneous–homogeneous super-bands that explicitly model spectral structure.
4) We pioneer incorporating quaternion algebra with intrinsic spectral principles in HAD, providing a new sparse representation approach for detection networks.

The subsequent sections are arranged as follows. Section II reviews the related research and highlights our improvements. Then, the detailed structure of the proposed SuperB-QSNet is demonstrated in Section III. Section IV displays the experimental results and analysis. Section V summarizes the whole article and looks forward to possible future work.

II. Related Work

A. Dimensionality Reduction of Spectral Bands

As the resolution of spectral sensors improves, the volume of HSI increases significantly, often involving hundreds of spectral bands. While this enhanced resolution provides richer information, it also introduces challenges such as high computational costs and degraded detector performance due to the massive data volume (dimensionality curse) [4], [5], [6]. To address these challenges, dimensionality reduction has become a critical step in HAD [28], [29]. Despite its importance, most studies rely on traditional algorithms, treating spectral compression as an afterthought rather than a carefully designed process. For instance, Wang et al. [28] integrated principal component analysis (PCA) into a tensor low-rank and sparse representation (TLRSR) model, extracting a subset of bands to reduce dimensionality while preserving essential features. Similarly, Li et al. [29] projected the original unfolded HSI data into kernel space, retaining only the first k principal components for subsequent processing. While these methods achieve dimensionality reduction, they treat bands as independent entities, ignoring the intrinsic spectral mechanisms of homogeneity and heterogeneity that can enable natural spectrum compression [23], [27]. As a result, they risk discarding subtle spectral signatures critical for detecting minor anomalies, which often occupy only a few pixels with minimal spectral deviations [30], [31]. Identifying the anomalies is challenging, and any loss of information during compression can further obscure them, leading to missed detections or false alarms. Although Yan et al. [32] selected the most relevant and informative bands after a coarse-to-fine grouping process, they still overlooked the correlations among these retained bands.

To address these limitations, we propose HH-SBC that explicitly respects the spectral structure of HSI, as shown in Fig. 1(a). The proposed method begins with a designed heterogeneous–homogeneous cross-segmentation algorithm, which extends the entropy rate-based superpixel segmentation technology [22] to the spectral domain and visually demonstrates the segmentation results. This technique organizes spectral bands into meaningful heterogroups based on their physical characterization rather than relying on simplistic clustering metrics [33]. As a result, bands within the same super-band exhibit significantly similar characteristics, ensuring alignment with the underlying spectral mechanisms. Subsequently, a parallel quantile-based band selection selects four representative bands per group, naturally reducing redundancy and retaining the representative spectral information for accurate anomaly detection.

B. Low-Rank Representation for HAD

While the obtained super-bands preserve spectral context, spatial contextual information is also crucial for HAD. Most

HAD models are grounded in the spatial-domain assumption that background pixels dominate the spatial scene statistically [14], [15], [34], [35]. Based on this assumption, anomalies are typically detected according to the deviations from the background reconstructed by artificial dictionaries. However, extracting a pure and comprehensive background dictionary is inherently challenging. Many methods adopt complex strategies to approximate the low-rank background [34], [35] or use global projecting approaches like PCA-based collaborative representation [15] and RPCA [14], which risks embedding anomalies into the background due to spectral similarity and leads to false negatives. This issue may be worsened by the loss of subtle spectral signatures during dimensionality reduction. Furthermore, these approaches fail to exploit cross-channel correlations within homogeneous spectral regions, which are the critical mechanisms for capturing consistent yet subtle spectral patterns and the unique advantages of high-dimensional HSIs.

Quaternions, which extend the concept of complex numbers with one real part and three imaginary parts, can fully encode the complex interspectral relationships while preserving the spatial structure of the original data. The quaternion transformer network (QTN) embeds the algebraic properties of quaternions into the Transformer architecture after band adaptive selection, simultaneously capturing long-range spatial–spectral dependencies in a hypercomplex space [36]. In [37], a quaternion wavelet transform matrix is proposed for HAD, combining an optimal clustering framework to reduce dimensionality and eliminate spectral redundancy. To adapt to the high-dimensional spectral characteristics of hyperspectral imagery, these algorithms significantly reduce the data dimensionality before quaternion representation, resulting in the loss of detailed high-resolution spectral information.

Therefore, we propose QSR, as shown in Fig. 1(b). In this framework, each super-band is encoded as a quaternion signal with one real and three imaginary components, inherently preserving the spatial structure and adapting to intragroup spectral correlations through hypercomplex algebra [38]. It focuses on intragroup relationships of homogeneous spectral bands rather than employing global processing of all bands, which may disrupt the original spectral structures and introduce unnecessary interference. While Tu et al. [39] combined quaternions to implement the parallel Fourier transform, they did not explore the domains of low rank and sparse decomposition. Our model integrates quaternion theory with intrinsic spectral principles into HAD for the first time. Moreover, it extracts potential anomalies and directly discards low-rank parts. The extracted sparse component does not need to be a pure or comprehensive representation of anomalies; instead, it filters out the majority of background pixels, providing a refined input that facilitates subsequent detection networks. By jointly leveraging intergroup heterogeneity and intragroup correlation, QSR can achieve sensitivity to subtle anomalies.

C. Deep Learning Detectors of HAD

Current deep learning HAD models often fail to adequately exploit the intrinsic structural properties of spectral data, primarily due to their inadequate modeling of spectral homogeneity and heterogeneity information [21], [40], [41]. When incorporating spatial priors to identify anomalies, existing methods typically process spectral bands as independent channels, neglecting the explicit modeling of internal discriminative structures inherent in spectral data. This manifests in two ways: methods either apply spatial operations independently to each spectral band without exploiting interband physical correlations or process all bands through global neural network layers that do not distinguish between homogeneous groups requiring joint processing and heterogeneous groups requiring differential treatment. For instance, while PDBSNet determines spectral characteristics through spatial neighborhood analysis and employs blind-spot reconstruction, it fundamentally treats spectral bands independently rather than exploiting their physical correlations and natural grouping [21]. Similarly, RGAE processes spectral dimensions through graph autoencoders but does not consider the complementary relationships between homogeneous and heterogeneous spectral channels that reflect underlying material properties [40].

To overcome these limitations, we propose DG-QCD that comprehensively leverages spatial and spectral structural priors, as shown in Fig. 1(c). In contrast to traditional networks that process spectral bands independently, our detector utilizes quaternion-based group processing to explicitly model cross-band interactions within homogeneous spectral groups while maintaining discriminative boundaries between heterogeneous groups. The detector synergizes with our HH-SBC and QSR components by consistently maintaining spectral grouping structure throughout the detection pipeline. It also amplifies discriminative features by computing differences between adjacent heterogeneous groups. This integrated framework guarantees continuous preservation of spectral structural integrity from initial dimensionality reduction to final detection through an efficient unsupervised approach.

III. Proposed SuperB-QSNet

In this article, bold straight letters represent 3-D data cubes, i.e., $\mathbf{X}$, bold capital italic letters denote 2-D matrices, i.e., $\boldsymbol{X}$, bold lowercase italic letters indicate 1-D vectors, i.e., $\boldsymbol{x}$, and scalars are presented in italic letters, i.e., x. In addition, the quaternion tensor and matrix are represented by capital and bold capital Greek letters, i.e., $\mathcal{X}$ and $\boldsymbol{\mathcal{X}}$, respectively.

A. Heterogeneous–Homogeneous Super-Band Compression

Real-world objects in HSIs exhibit spatial continuity, forming clustered pixel groups rather than scattered distributions. This makes superpixel segmentation ideal for remote sensing as it: 1) groups similar pixels into homogeneous regions and 2) preserves object boundaries through adaptive region growing [42]. The entropy rate method enhances this by optimizing compactness and connectivity to generate more regular and content-sensitive superpixels [22]. However, current superpixel methods have a limitation for high-dimensional data [15]; they require reducing the hyperspectral data to grayscale, which loses valuable spectral information. Critically, HSIs exhibit an analogous coherence in the spectral dimension, driven by the physical properties of materials that adjacent bands often capture correlated reflectance patterns from the same material. Just as superpixels exploit local pixel similarities to generate meaningful homogeneous regions (where each region typically corresponds to a distinct object), this spectral coherence suggests that the global spectral bands could be segmented into homogeneous spectral units. Therefore, we can partition the HSI cube into spectrally contiguous regions while retaining the spatial structure, providing the foundation for downstream tasks.

We extend the entropy rate superpixel segmentation [22] to the spectral domain, yielding regular and content-sensitive groups for natural spectral compression, preserving primary spectral structure and reducing redundancy. It also enables intuitive visualization and enhances the interpretability of spectral variations. The critical challenge in extending this methodology to the spectral dimension lies in two aspects [42]. One is to minimize pixel quantity to emphasize spectral variations over spatial effects. The segmentation process becomes compromised when incorporating multiple pixels, as even two distinct pixels can distort heterogeneous band segmentation. This occurs because the resulting edges can capture both spectral differences and spatial variations, making it challenging to distinguish spectral heterogeneity. The other is that segmentation can only directly operate on a suitable 2-D structural matrix, even if the spectrum lacks such attributes.

To address these issues, we design a heterogeneous–homogeneous cross-segmentation algorithm to generate robust super-bands. Assume that the original HSI can be presented as a 3-D tensor $\mathbf{H} \in \mathbb{R}^{r\times c\times b}$ where r, c, and b are the numbers of spatial rows, spatial columns, and spectral bands, respectively. By unfolding the 2-D spatial structure, the 3-D tensor $\mathbf{H}$ can be reshaped as $\boldsymbol{H} = [\boldsymbol{h}_1, \boldsymbol{h}_2, \ldots, \boldsymbol{h}_i, \ldots, \boldsymbol{h}_s] \in \mathbb{R}^{s\times b}$, where $s = r \times c$ and $\boldsymbol{h}_i \in \mathbb{R}^{s\times 1}$ is the ith $(i = 1, 2, \ldots, b)$ spectral vector and s means the total number of pixels. Besides, the global band vector of the jth pixel can be represented as $\boldsymbol{p}^j \in \mathbb{R}^{b\times 1}$. Thus, the process of heterogeneous–homogeneous segmentation is displayed as follows.

1) Segmentation on a Single Spectral Matrix: It begins with randomly selecting an anchor pixel with the global band vector $\boldsymbol{p}^k$. Through column-wise replication of this vector ϖ times, we construct an operable 2-D spectral matrix $\boldsymbol{M}$ [27] as

$$\boldsymbol{M} = \left[\boldsymbol{p}^k, \boldsymbol{p}^k, \ldots, \boldsymbol{p}^k\right] \in \mathbb{R}^{b\times\varpi}. \tag{1}$$

The factor ϖ operates within a constrained parameter space $\varpi \in \mathbb{Z}^+$, where lower and upper bounds present distinct tradeoffs. If $\varpi = 1$, the system degenerates to a 1-D configuration, while $\varpi \gg 1$ exponentially increases the partitioning complexity. Our empirical analysis determined that $\varpi = 2$ represents the solution satisfying dimension requirements and computational efficiency [27]. The reconstruction of $\boldsymbol{M}$ mitigates pixel-level differences and establishes the structural framework for subsequent super-band generation. Then, we employ entropy rate-based segmentation [15] on $\boldsymbol{M}$ to obtain an edge subset $\boldsymbol{t}$ by

$$\underset{\boldsymbol{t}}{\operatorname{argmax}}\left(\operatorname{tr}\left(R\left(\boldsymbol{t}\right)\right) + \alpha B\left(\boldsymbol{t}\right)\right) \tag{2}$$

where $\operatorname{tr}(\cdot)$, $\mathrm{R}(\cdot)$, and $\mathrm{B}(\cdot)$ refer to the trace function, entropy rate term, and balancing term with a balancing parameter α, respectively. As a result, the obtained heterogeneous edges can be directly located between adjacent bands, completely segmenting heterogeneous band groups. We denote the initial index of each super-band as the heterogeneous edge position. Given the number $n-1$ of heterogeneous edges and the initial band index $t_1 = 1$ of the first super-band, the boundary position can be indicated as $\boldsymbol{t} = \{t_1, t_2, \ldots, t_n\}$. However, segmentation on a single spectral matrix has a noteworthy limitation. Its performance relies heavily on the random selection of the anchor pixel and its spectral observation vector $\boldsymbol{p}^k$, which can result in inconsistent spectral clustering outcomes across different executions. Therefore, we develop heterogeneous–homogeneous cross-segmentation with double medians.

2) Cross-Segmentation With Double Median: In the cross-segmentation, we employ a multitrial consensus strategy to mitigate instability from random anchor pixel selection. The position indexes of heterogeneous edges are first aggregated within each trial via median filtering to suppress outlier effects, followed by a second median filtering across all trials to eliminate dependence on random initialization. Specifically, we first conduct five independent trials to establish reliable spectral boundaries, ensuring statistical reliability. For each trial m $(m = 1, 2, \ldots, 5)$, we randomly select ten distinct anchor pixels to construct corresponding spectral matrices $\boldsymbol{M}_i^{(m)} \in \mathbb{R}^{b\times\varpi}$ $(i = 1, 2, \ldots, 10)$ by (1). Then, the index sets of heterogeneous edge positions $\boldsymbol{t}_i^{(m)} \in \mathbb{R}^{n\times 1}$ are obtained by conducting (2) on each spectral matrix. For each trial, the within-trial median is computed position-wise across the ten vectors

$$\boldsymbol{t}^{(m)} = \operatorname{median}\left(\boldsymbol{t}_1^{(m)}, \boldsymbol{t}_2^{(m)}, \ldots, \boldsymbol{t}_{10}^{(m)}\right). \tag{3}$$

Here, the jth $(j = 1, 2, \ldots, n)$ element of $\boldsymbol{t}^{(m)}$ is the median of the jth boundary position across all ten anchor pixels in trial m. After repeating the above for five trials and sorting the position indices in ascending order, holding the arrangement structure of $n \times 5$ to eliminate fluctuations and even significant variations across different trials, the final heterogeneous edge indexes are computed by taking the cross-trial median for each position j across the five trials

$$\boldsymbol{t}^* = \operatorname{median}\left(\boldsymbol{t}^{(1)}, \boldsymbol{t}^{(2)}, \ldots, \boldsymbol{t}^{(5)}\right) = \left[c_1^*, c_2^*, \ldots, c_n^*\right]^{\mathrm{T}}. \tag{4}$$

Using the stabilized edge indexes c_1^*, c_2^*,..., c_n^*, we partition the original hyperspectral data into heterogeneous–homogeneous super-bands as

$$\boldsymbol{S}_j = \left\{\boldsymbol{h}_{c_j^*},\ \boldsymbol{h}_{c_j^*+1},\ \ldots,\ \boldsymbol{h}_{c_{j+1}^*-1}\right\}, \quad j = 1, 2, \ldots, n. \tag{5}$$

Here, each super-band $\boldsymbol{S}_j$ represents a spectrally coherent region, ensuring meaningful spectral grouping. The whole procedure of heterogeneous–homogeneous cross-segmentation is displayed in Algorithm 1.

Fig. 2 presents the cross-segmentation results before and after each step of the double median filtering. The yellow lines on each matrix mark the positions of heterogeneous edges acquired by a single spectral matrix. The red line within each trial and the blue line across trials represent the median edge indices produced by the first intragroup

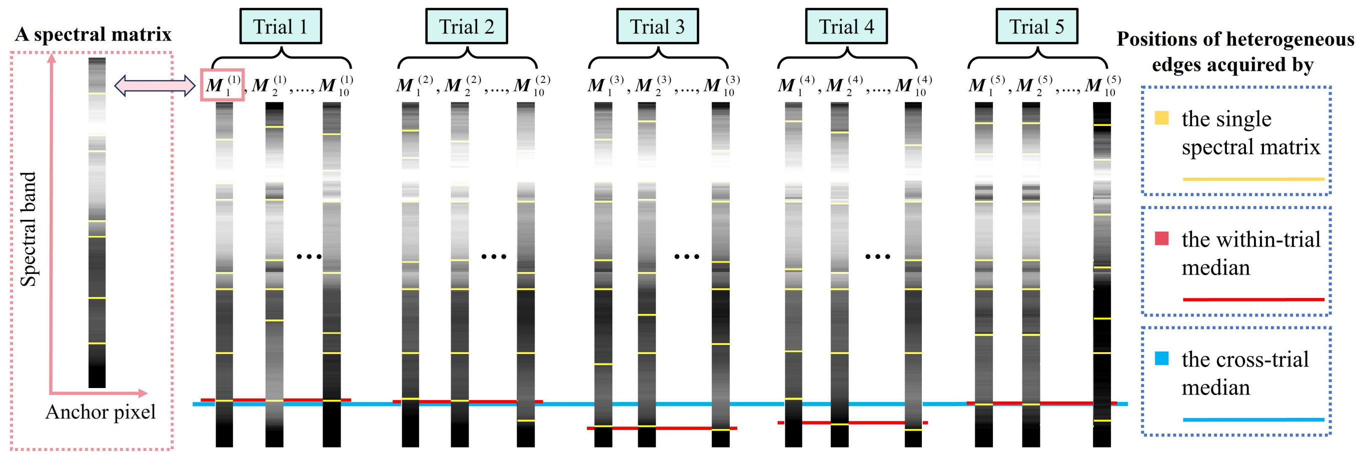

Fig. 2. Cross-segmentation results before and after each step of the double median approach.

Algorithm 1 Cross-Segmentation With Double Median

Input: $\mathbf{H} \in \mathbb{R}^{r\times c\times b}$, anchor pixel number per trial i, independent trial number m, target edge count n
Output: Super-bands $\{\boldsymbol{S}_j\}_{j=1}^{n}$
Initialize: Final edge indexes $\boldsymbol{t}^* = \emptyset \in \mathbb{R}^{n\times 1}$
for $m = 1$ to 5 **do**
1: Randomly select 10 distinct anchor pixels
for $i = 1$ to 10 **do**
1.1: Construct matrices $\boldsymbol{M}_i^{(m)} \in \mathbb{R}^{b\times\varpi}$ via (1)
1.2: Obtain edges $\boldsymbol{t}_i^{(m)} \in \mathbb{R}^{n\times 1}$ via (2)
end for
2: Compute within-trial median $\boldsymbol{t}^{(m)}$ via (3)
3: Sort and store: $\boldsymbol{t}^*[:, m] \leftarrow \boldsymbol{t}^{(m)}$
end for
4: Compute cross-trial median $\boldsymbol{t}^* \leftarrow [c_1^*, c_2^*, \ldots, c_n^*]^{\mathrm{T}}$ via (4)
for the super-band number $j = 1$ to n **do**
5: Extract super-band $\mathcal{S}_j$ via (5)
end for

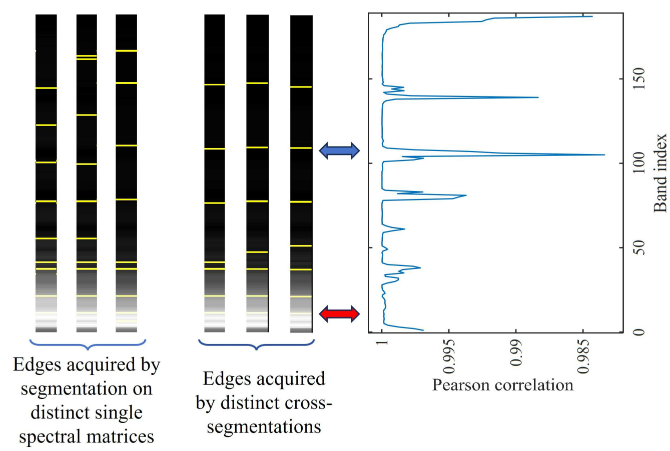

Fig. 3. (Left) Edges acquired by segmentation on distinct single spectral matrices. (Middle) Edges acquired by distinct cross-segmentations. (Right) Pearson correlation curve between adjacent spectral bands.

and the second intergroup median filtering, respectively. As shown in Fig. 2, the edges on individual 2-D spectral matrices exhibit significant variations as the anchor points change. In contrast, the within-trial median buffers this variability, and the cross-trial median further enhances the robustness of the final results.

Fig. 3 demonstrates the heterogeneous edges derived from the segmentation of separate spectral matrices and independent cross-segmentations, along with similarity curves in Pearson correlation between adjacent spectral bands in the Cat Island dataset. Notably, the edges from cross-segmentation are more precise and align better with the similarity curve's troughs than those from individual spectral matrix segmentation. The blue double arrow highlights areas (upper portion) where both methods successfully detected spectral differences, reflecting significant fluctuations in the correlation coefficient curve. Conversely, the red double arrow (lower portion) indicates the heterogeneous band edge not detected by the correlation coefficient curve on the right but detected by the visual spectral matrix on the left. This is probably because cross-segmentation enhances sensitivity to localized spectral variations with anchor pixels. Moreover, this method enhances the visual representation of spectral homogeneity and improves interpretability by resolving fine-grained spectral transitions that traditional metrics fail to detect.

3) Parallel Quantile-Based Band Selection: After cross-segmentation, we use a parallel quantile-based band selection technique to obtain four representative bands from each super-band. On the jth super-band $\boldsymbol{S}_j$, the simplified super-band is defined as

$$\boldsymbol{B}_j = \left\{\boldsymbol{q}_{0.3}^j, \boldsymbol{q}_{0.5}^j, \boldsymbol{q}_{0.7}^j, \boldsymbol{q}_{0.9}^j\right\}$$
$$\text{s.t.} \begin{cases} h = \left(c_{j+1}^* - c_j^* - 1\right) p + c_j^* \\ \boldsymbol{q}_p^j = \boldsymbol{h}_{\lfloor h\rfloor} + (h - \lfloor h\rfloor)\cdot\left(\boldsymbol{h}_{\lfloor h\rfloor+1} - \boldsymbol{h}_{\lfloor h\rfloor}\right). \end{cases} \tag{6}$$

Here, the spectral vector $\boldsymbol{q}_p^j$ corresponds to the pth quantile of the band index within the jth super-band. $\lfloor h\rfloor$ means to round down the quantile position h to the nearest integer index. The parallel quantile-based band selection achieves robust spectral compression through synergistic processing. First, the double median operation provides initialization invariance by aggregating results from multiple independent trials, eliminating dependence on any random anchor point. Second, the optimized quantile spacing ensures a comprehensive spectrum representation, specifically the 0.3th, 0.5th, 0.7th, and 0.9th quantiles. The median, i.e., 0.5th quantile, is a stable central reference, while the 0.3th and 0.7th quantiles bound the characteristic distribution, and the 0.9th quantile preserves critical boundary transitions. This synergistic approach, grounded in nonparametric statistics, addresses the sensitivity to uncertain random anchor pixels, reducing redundancy and enhancing discriminative features for anomaly detection.

B. QSR Model

Quaternion representation provides an ideal framework for hyperspectral data analysis by fundamentally preserving multiband spectral relationships through its hypercomplex algebraic structure. Unlike conventional real-valued methods that process spectral bands independently, quaternion algebra inherently maintains the natural coupling between spectrally homogeneous bands via its 4-D representation [38]. The hypercomplex domain provides a compact unified representation that processes multiple similar bands as single algebraic entities, thereby eliminating spectral discontinuities introduced by traditional band-wise processing while significantly reducing computational complexity. Our QSR model capitalizes on this advantage by processing each spectrally homogeneous super-band as an indivisible quaternion unit rather than isolated bands. This approach offers the critical benefits of eliminating artificial spectral distortions caused by global band processing and removing the need for explicit background modeling through built-in noise suppression. QSR also retains the original characteristics of spectral homogeneity and heterogeneity for downstream analysis during the input–output transformation, preventing the information loss often associated with real-valued decomposition methods.

Through reasonable super-band quantity settings, we can achieve that each simplified super-band $\boldsymbol{B}_j$ $(j = 1, 2, \ldots, n)$ contains four spectral band vectors, creating conditions for quaternion representation. By parallel folding each matrix $\boldsymbol{B}_j \in \mathbb{R}^{s\times 4}$ to the tensor $\mathbf{B}_j = \mathrm{fold}(\boldsymbol{B}_j) = \{\boldsymbol{Q}^j_{0.3}, \boldsymbol{Q}^j_{0.5}, \boldsymbol{Q}^j_{0.7}, \boldsymbol{Q}^j_{0.9}\}$ following the spatial structure, the matching quaternion transformation $q(\mathbf{B}_j)$ can be presented as:

$$\mathcal{B}^j = q\left(\mathbf{B}_j\right) = \boldsymbol{Q}^j_{0.3} + \boldsymbol{Q}^j_{0.5}\phi + \boldsymbol{Q}^j_{0.7}\varphi + \boldsymbol{Q}^j_{0.9}\gamma \tag{7}$$

where $\mathcal{B}^j \in \mathbb{Q}^{r\times c}$ $(j = 1, 2, \ldots, n)$ are generated quaternion matrices and $\{1, \phi, \varphi, \gamma\}$ are the basis units of any quaternions [27]. The quaternion framework provides an effective mechanism for joint spatial–spectral representation by inherently preserving cross-channel correlations through its hypercomplex algebra structure. We implement a targeted quaternization process for each super-band to leverage this property.

Unlike conventional detection methods [29] that depend on artificially constructed background dictionaries, our approach introduces a paradigm-shifting solution through quaternion-based sparse recovery on each super-band. Specifically, the core optimization problem of QSR on $\mathcal{B}^j \in \mathbb{Q}^{r\times c}$ $(j = 1, 2, \ldots, n)$ is formulated as follows:

$$\min_{\mathcal{A}^j,\mathcal{N}^j} \|\mathcal{A}^j\|_{2,1} + \lambda^j\|\mathcal{N}^j\|_*, \quad \text{s.t.} \quad \mathcal{B}^j = \mathcal{A}^j + \mathcal{N}^j \tag{8}$$

where λ^j, $\mathcal{A}^j$, and $\mathcal{N}^j$ are the regularization parameters referring to the jth component proportion, the anomaly quaternion, and the low-rank part derived from the jth super-band B_j, respectively. Besides, $\|\cdot\|_{2,1}$ is the $\ell_{2,1}$-norm for joint sparsity in a quaternionic complex space, and $\|\mathcal{N}^j\|_* = \sum\sigma(\mathcal{N}^j)$ is the quaternion nuclear norm on $\mathcal{N}^j$ with singular value operation $\sigma(\cdot)$. The $\ell_{2,1}$-norm $\|\cdot\|_{2,1}$ on $\mathcal{A}^j$ equals $\sum_{\mu=1}^{r}\left(\sum_{\nu=1}^{c}|\mathcal{A}^j(\mu,\nu)|^2\right)^{1/2}$, where $|\cdot|$ is the modulus operator and (μ, ν) is the spatial position of any pixel [38].

Algorithm 2 QSR on $\mathcal{B}^j$

Input: $\mathcal{B}^j$, λ, $\rho = 1.25$, $\varepsilon = 10^{-7}$
Output: Sparse anomaly part $\mathbf{A}^j \in \mathbb{R}^{r\times c\times 4}$
Initialize: $\mu_0 = 1.25/\max(\sigma(\mathcal{B}^j))$, $\mathcal{N}^j_0 = \hat{0} \in \mathbb{Q}^{r\times c}$, $t = 0$, $\mathcal{Y}^j_0 = \mathcal{B}^j/\max(\max(\sigma(\mathcal{B}^j)), \lambda^2\max(|\mathcal{B}^j(\mu,\nu)|))$
while not converged **do**
1: $t = t + 1$
2: Update $\mathcal{A}^j$ by
$\mathcal{A}^j_t = \arg\min_{\mathcal{A}^j}\left(\left\|\mathcal{A}^j\right\|_{2,1} + \frac{\mu_{t-1}}{2}\left\|\mathcal{N}^j_{t-1} + \mathcal{A}^j - \mathcal{B}^j + \frac{\mathcal{Y}^j_{t-1}}{\mu_{t-1}}\right\|^2_{\mathrm{F}}\right)$
3: Update $\mathcal{N}^j$ by
$\mathcal{N}^j_t = \arg\min_{\mathcal{N}^j}\left(\left\|\mathcal{N}^j\right\|_* + \frac{\mu_{t-1}}{2}\left\|\mathcal{N}^j + \mathcal{A}^j_t - \mathcal{B}^j + \frac{\mathcal{Y}^j_{t-1}}{\mu_{t-1}}\right\|^2_{\mathrm{F}}\right)$
4: $\mathcal{Y}^j_t = \mathcal{Y}^j_{t-1} + \mu_{t-1}(\mathcal{B}^j - \mathcal{A}^j_t - \mathcal{N}^j_t)$
5: $\mu_t = \rho\mu_{t-1}$
Check the convergence conditions (9)
end while
6: Perform the inverse $\mathrm{q}^{-1}(\cdot)$ of transformation (7)
$\mathbf{A}^j = \mathrm{q}^{-1}(\mathcal{A}^j)$

To address the optimization problem and derive the optimal solution, we utilize an inexact augmented Lagrange multiplier [39], which provides guaranteed convergence properties. Empirical evidence shows that the algorithm generally converges within 20–30 iterations for standard hyperspectral datasets, primarily influenced by the employed quaternion singular value decomposition computation. The convergence criterion monitors the relative primal residual by

$$\frac{\|\mathcal{B}^j - \mathcal{A}^j - \mathcal{N}^j\|_F}{\|\mathcal{B}^j\|_F} < \varepsilon \tag{9}$$

where the quaternionic Frobenius norm on $\mathcal{B}^j$ is expressed as $\|\mathcal{B}^j\|_F = \left(\sum_{\mu=1}^{r}\sum_{\nu=1}^{c}|\mathcal{B}^j(\mu,\nu)|^2\right)^{1/2}$. The whole procedure to obtain the sparse anomaly $\mathcal{A}^j$ and the corresponding real $\mathbf{A}^j \in \mathbb{R}^{r\times c\times 4}$ is detailed in Algorithm 2.

Fig. 4 displays the decomposition results of QSR applied to San Diego, showcasing bands from the 9th and 18th super-bands. The original bands (top row) contain mixed information, while the sparse components (middle row) isolate distinctive spectral signatures representing anomalous. By comparing red-highlighted regions across gray images, it is

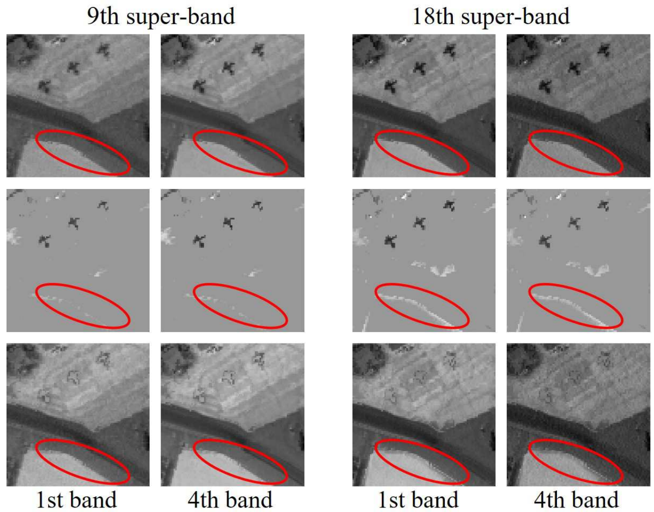

Fig. 4. Grayscale images of different bands within distinct super-bands. (First to third rows) Original band, the retained sparse components, and the discarded low-rank components.

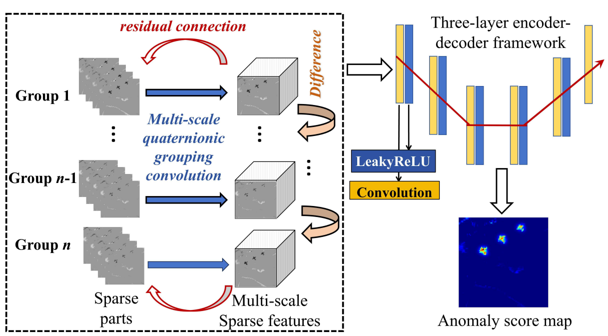

Fig. 5. Flowchart of the proposed DG-QCD.

clear that these sparse components maintain consistent patterns within each super-band while preserving the differences between super-bands, demonstrating how QSR effectively captures spectral intragroup homogeneity and intergroup heterogeneity. The decomposition quality is particularly evident in the transition areas between different surface materials, where the sparse components clearly emphasize boundary features. However, the sparse components still contain traces of background elements, and the low-rank components (bottom row) retain some anomalous features. This overlap between components confirms a fundamental challenge in matrix decomposition; perfect separation is rarely achievable. Despite this limitation, the quaternionic model offers an advantage by concentrating most of the spectrally significant information in the sparse components, thereby providing reliable input for the downstream detector.

C. Difference-Guided Quaternionic Convolution Detector

Unlike conventional methods that rely on low-rank components or original data for background reconstruction, our approach completely discards the low-rank component and solely depends on the sparse component. Crucially, the retained sparse elements preserve the key super-band properties of intragroup homogeneity and intergroup heterogeneity. To amplify the contrast between sparse anomalies and the remaining background in each $\mathbf{A}^j$, an unsupervised reconstruction module DG-QCD is designed, as illustrated in Fig. 5.

1) Multiscale Quaternionic Grouped Convolution: First, a multiscale quaternionic grouped convolution is performed on each homogeneous group $\mathbf{A}^j \in \mathbb{R}^{r\times c\times 4}$ obtained by QSR as

$$\mathbf{G}_\delta\left(\mathbf{A}^j\right) = \mathbf{A}^j \circledast \mathbf{W}_\delta^{(\mathrm{quat})}, \quad \delta \in \{1,3,5,7\} \tag{10}$$

where $\mathbf{G}_\delta(\cdot)$ is a quaternionic multiscale operation with the grouped convolution operator $\circledast$, the grouped convolution kernel $\mathbf{W}_\delta^{(\mathrm{quat})} \in \mathbb{R}^{C_\delta\times 4\times\delta\times\delta}$, and the size of the quaternion convolution kernel δ. It preserves local relationships among homogeneous bands through grouped execution rather than single-band traversal. In this step, we do not implement actual quaternion calculations. Instead, inspired by the parallel quaternion framework introduced in Section III-B, we group the pre-divided "quaternion" spectral bands and feed them directly into the convolution network as independent units. This approach eliminates redundant computational steps while preserving the structural advantages of the quaternion-inspired design. After employing the quaternionic grouping architecture that includes all kernel sizes, the feature maps $\mathbf{G}_\delta(\mathbf{A}^j) \in \mathbb{R}^{r\times c\times C_\delta}$ of different scales are connected to $\mathbf{T}^j \in \mathbb{R}^{r\times c\times C_T}$ by

$$\mathbf{T}^j = \mathrm{Concat}\left(\mathbf{G}_1\left(\mathbf{A}^j\right), \mathbf{G}_3\left(\mathbf{A}^j\right), \mathbf{G}_5\left(\mathbf{A}^j\right), \mathbf{G}_7\left(\mathbf{A}^j\right)\right) \tag{11}$$

where Concat$(\cdot)$ stands for connect features of multiple scales along the spectral dimension and $C_T = C_1 + C_3 + C_5 + C_7$. In our implementation, we set $C_1 = C_3 = C_5 = C_7 = C_T/4 = 16$ to ensure balanced multiscale representation. This multiscale feature extraction method allows the model to have more substantial detection capabilities for anomaly targets of various sizes. However, such networks easily yield gradient disappearance problems, decreasing feature expression ability. To solve this problem, a residual connection mechanism is employed to obtain $\mathbf{Y}^j \in \mathbb{R}^{r\times c\times C_T}$ as

$$\mathbf{Y}^j = \mathbf{T}^j + \mathcal{P}_{1\times 1}\left(\mathrm{Concat}\left(\mathbf{T}^{j-1}, \mathbf{T}^j, \nabla^j\right)\right) \tag{12}$$

where $\mathcal{P}_{1\times 1}(\cdot) : \mathbb{R}^{r\times c\times 3C_T} \rightarrow \mathbb{R}^{r\times c\times C_T}$ is a 1×1 group connection layer to ensure dimensional consistency for all residual connections. ∇^j represents the computed intergroup gradient features. This residual connection mechanism ensures that the input information can be directly passed to the output, effectively alleviating the gradient vanishing problem and enhancing expression capabilities.

2) Difference-Guided Anomaly Detection: Since bands from heterogeneous super-groups often present different anomaly reflectance values, we further calculate the tensor difference $\mathbf{D}^j$ between the $(j-1)$th and jth super-bands by

$$\mathbf{D}^j = \mathbf{Y}^j - \mathbf{Y}^{j-1}, \quad j \in \{2,3,\ldots,n\}. \tag{13}$$

This difference between groups can effectively capture the band change mode and is significant for detecting spectral anomalies. By calculating the absolute difference of the characteristics of adjacent groups, the model can pay attention to areas with abnormal relationships between bands.

Then, the proposed detector connects the attributes of all super-bands and the differences between groups through a three-layer encoder–decoder framework. The encoder part includes group convolution, multiscale features, and group differences

$$\begin{aligned}\mathbf{E}_1 &= \text{LeakyReLU}\left(G_3\left(\text{Concat}\left(\mathbf{Y}^1,\ldots,\mathbf{Y}^n,\mathbf{D}^2,\ldots,\mathbf{D}^n\right)\right)\right)\\ \mathbf{E}_2 &= \text{LeakyReLU}\left(G_3\left(\mathbf{E}_1\right)\right)\\ \mathbf{E}_3 &= \text{LeakyReLU}\left(G_3\left(\mathbf{E}_2\right)\right)\end{aligned}\tag{14}$$

where $\text{LeakyReLU}(x) = \max(0.2x, x)$ is an activation function. This part can help the model learn a compact data representation, removing redundant information. The three-layer decoder then tries to rebuild the compressed features back to $\mathbf{K}$, holding the original dimension of the global sparse component $\mathbf{A} = \text{Concat}(\mathbf{A}^1, \mathbf{A}^2, \ldots, \mathbf{A}^n) \in \mathbb{R}^{r\times c\times 4n}$ by

$$\begin{aligned}\mathbf{J}_1 &= \text{LeakyReLU}\left(G_3\left(\mathbf{E}_3\right)\right)\\ \mathbf{J}_2 &= \text{LeakyReLU}\left(G_3\left(\mathbf{J}_1\right)\right)\\ \mathbf{K} &= G_3\left(\mathbf{J}_2\right).\end{aligned}\tag{15}$$

The initial anomaly score map is derived from the pixel-wise mean square error between the sparse component $\mathbf{A}$ and its reconstruction $\mathbf{K}$

$$\boldsymbol{L}_{\text{base}}\left(\mathbf{A},\mathbf{K}\right) = \frac{1}{4n}\sum_{i=1}^{4n}\left(\mathbf{A}\left(:,:,i\right) - \mathbf{K}\left(:,:,i\right)\right)^2 \tag{16}$$

where $\boldsymbol{L}_{\text{base}}$ is the base reconstruction error map, and the summation is performed over the $4n$ channels of the concatenated sparse component.

Rather than directly utilizing this basic reconstruction error, we introduce a novel reconstruction quality-aware loss (RQAL) that dynamically modulates the optimization process based on the intrinsic reconstructability of different image regions. This approach is grounded in the key observation that the sparse component $\mathbf{A}$ can be divided into easily reconstructible background regions (characterized by near-zero values and minimal spectral variation) and challenging anomalous regions (exhibiting higher energy and spatial–spectral deviation). RQAL operates through two fundamental computational stages. First, it dynamically generates a quality mask by thresholding the instantaneous pixel-wise reconstruction error

$$\boldsymbol{M}_{\text{high}} = \mathcal{M}\left(\boldsymbol{L}_{\text{base}}\left(\mathbf{A},\mathbf{K}\right) < \tau\right) \tag{17}$$

where $\tau = 0.001$ denotes the quality threshold, $\mathcal{M}(\cdot)$ is the indicator function that returns 1 for true conditions and 0 otherwise, and $\boldsymbol{M}_{\text{high}}$ is the resulting binary mask identifying well-reconstructed pixels, which correspond primarily to the background regions of $\mathbf{A}$. The final training loss $\mathcal{L}_{\text{RQAL}}$ is then computed as a selectively masked average focused exclusively on these high-quality regions

$$\mathcal{L}_{\text{RQAL}} = \frac{\sum_{i,j}\left[\boldsymbol{L}_{\text{base}}\left(\mathbf{A},\mathbf{K}\right)_{i,j}\cdot\boldsymbol{M}_{\text{high}_{i,j}}\right]}{\sum_{i,j}\boldsymbol{M}_{\text{high}_{i,j}} + \epsilon} \tag{18}$$

where the indices i and j iterate over all spatial positions in the error map and ϵ is a small constant (e.g., 1×10^{-8}) included for numerical stability to prevent division by zero.

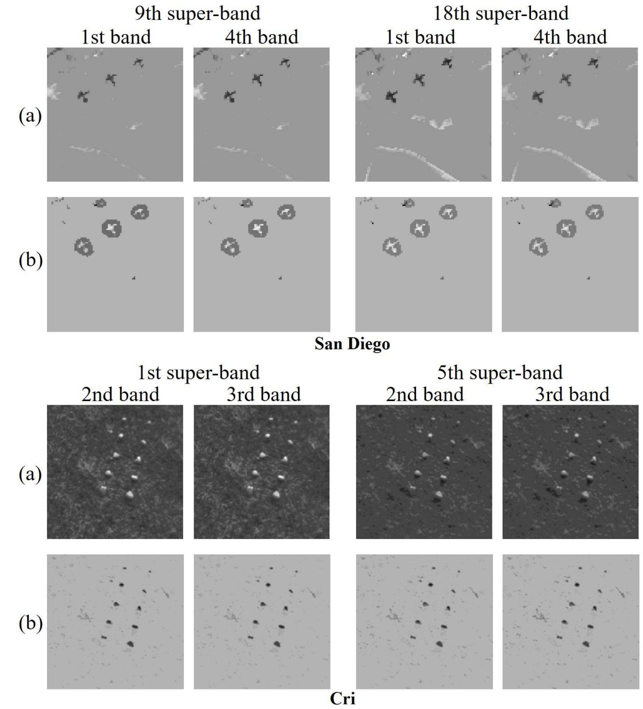

Fig. 6. Grayscale images of sparse components obtained by (a) QSR and (b) DG-QCD on different bands within distinct super-bands on San Diego and Cri.

The encoder–decoder network is trained in a fully unsupervised manner for around 50 epochs using the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and an initial learning rate of 1×10^{-4}. This formulation ensures that gradient updates explicitly enhance the model's capacity to reconstruct the predominant background patterns while strategically ignoring anomalous regions. Consequently, the model progressively refines its background reconstruction capability, thereby amplifying the discrimination between background and anomalies in the final anomaly score. For inference, the normalized anomaly score map is obtained by min–max scaling the base reconstruction error $\boldsymbol{L}_{\text{base}}(\mathbf{A},\mathbf{K})$ to the range [0, 1], where higher values indicate a greater probability of anomaly presence.

Fig. 6 presents the grayscale images of sparse components obtained by (a) QSR and (b) DG-QCD on different bands within distinct super-bands on San Diego and Cri datasets. This visualization demonstrates the progressive refinement of anomaly components through our proposed pipeline. As illustrated in Fig. 6, the sparse components extracted by QSR contain valuable anomaly information but exhibit fragmented and scattered spatial patterns across different spectral bands. While QSR successfully decomposes the hyperspectral data into sparse and low-rank components, isolating potential anomalous pixels from the background subspace, the resulting sparse components lack spatial coherence due to the pixel-wise nature of the sparse decomposition process. The anomaly components obtained through DG-QCD demonstrate the effectiveness of the subsequent encoder–decoder refinement stage. After processing the QSR-generated sparse components, DG-QCD produces considerably more precise and concentrated spatial distributions. In the refined anomaly component maps, anomalous targets appear as white regions on San Diego and dark regions on Cri, featuring well-defined boundaries and regular shapes with strong contrasts against the gray background. This transformation demonstrates effective spatial aggregation and enhanced visibility of anomalous information.

IV. Experiments

All the experiments are primarily performed on a laptop with a 2.60-GHz CPU running on a Windows 11 system, Intel Core i9-13900H processors, and 32 GB of memory.

A. Experiment Datasets

To quantitatively verify the proposed model's performance, we conducted experiments on the following six publicly available real-world HSIs. The false-color images and the ground-truth maps are given in Fig. 7(a) and (b).

1) *San Diego:* This dataset, acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over San Diego, CA, USA, consists of 189 spectral bands after excluding noisy and water absorption bands. It has 90×90 pixels with 3.5-m spatial resolution, where 77 pixels are identified as anomaly targets, representing 0.95% of the total [43]. The targets are in three relatively focused areas.
2) *Cri:* Collected by the Nuance Cri hyperspectral sensor over Wuhan University, China, this dataset contains 46 spectral channels retained at 10-nm spectral resolution and 400×400 pixels. There are 860 anomalies, accounting for 0.54% of total pixels [44]. The targets are in ten relatively scattered areas.
3) *Bay Champagne:* Captured by AVIRIS, this dataset contains 188 spectral bands (after preprocessing) and 100×100 pixels, with a spatial resolution of 4.4 m/pixel. The scene includes 11 anomalous pixels occupying 0.11% of the global space [45]. The targets are concentrated in a small area.
4) *Texas Coast:* Acquired by AVIRIS along the Texas Coast (Texas-C), USA, this dataset consists of 207 spectral bands (after removing absorption and low-signal-to-noise ratio (SNR) bands) and 100×100 pixels with 17.2-m spatial resolution. This scene contains 155 pixels identified as anomaly targets, representing 1.55% of the total [46]. The targets are in 20 relatively focused areas.
5) *Pavia:* Acquired by the Reflectance Optical System Imaging Spectrometer (ROSIS-03) sensor over Pavia, Italy, this dataset comprises 102 spectral bands and 150×150 pixels, with 1.3-m spatial resolution. The anomalies include 68 target pixels (0.30% of the total), primarily vehicles on a bridge [47]. The targets are in seven relatively scattered areas.
6) *Cat Island:* Collected by AVIRIS around Cat Island, this dataset contains 188 spectral bands (after noise removal) and 150×150 pixels, with 17.2-m spatial resolution. The scene includes 19 anomalous pixels (0.08% of the total) corresponding to aircraft targets [46]. The targets are concentrated in a small area.

B. Experiment Settings

They are given as follows.

1) Evaluation Indexes: Multiple quantitative and qualitative metrics are employed to assess the performance.

1) *Anomaly Score Map:* It qualitatively and visually represents the effects of background suppression and highlights anomalies. By observing color changes on the maps, we can quickly identify anomalies.
2) *Receiver Operating Characteristic:* The qualitative receiver operating characteristic (ROC) curve graphically represents the tradeoff between the probability of detection (PD) and false alarm rate (FAR) across varying thresholds. The ROC curve plots the PD in the y-axis against the FAR in the x-axis, where curves closer to the upper left corner demonstrate superior detection performance, achieving higher detection rates while maintaining lower false alarm probabilities.
3) *AUC, Running Time, and MRs:* To thoroughly assess detection performance, we utilize three key quantitative metrics: area under the ROC curve (AUC), running time, and memory requirements (MRs). AUC measures the area under the ROC curve, where a larger value indicates greater detection accuracy. Additionally, we include running time and MR to evaluate the computational efficiency and space needed by the model, respectively.
4) *Separability Box Plot:* It is a statistical tool to qualitatively evaluate the distinction between the background and anomalies. In this method, distinct colored boxes represent the background and the anomaly parts. The greater the distance between the two boxes, the easier it is to differentiate the two parts.

2) Comparison Methods: Comprehensive comparisons are conducted by introducing various state-of-the-art HAD techniques, including statistical method FrFE-RX [12], representation-based methods comprising matrix-derived approaches, i.e., SuperRPCA [15] and low-rank and sparse representation (LRASR) [43], and tensor-based methods, i.e., GNLTR [14], PCA-TLRSR [28], and prior-based tensor approximation (PTA) [13], and deep learning methods, i.e., PDBSNet [21] and RGAE [40]. Among these methods, SuperRPCA and RGAE employ superpixel segmentation to capture homogeneous and heterogeneous spatial mechanisms, thereby enhancing spatial context extraction. Meanwhile, most experiments are conducted using the public codes from the related authors. However, due to discrepancies in dataset configurations, some methods lacked recommended parameter settings for our specific experimental setup. In such cases, we carefully adjusted the parameters to ensure proper execution.

C. Comparison of Performance

This section presents quantitative and qualitative comparison results on six benchmark hyperspectral datasets.

1) Anomaly Score Map: Fig. 7(c)–(k) presents the detection results in heatmap format across six original HSIs, comparing the performance of all nine detectors. Brighter pixels indicate higher anomaly probabilities in these visualizations, while darker regions represent background areas. It is worth

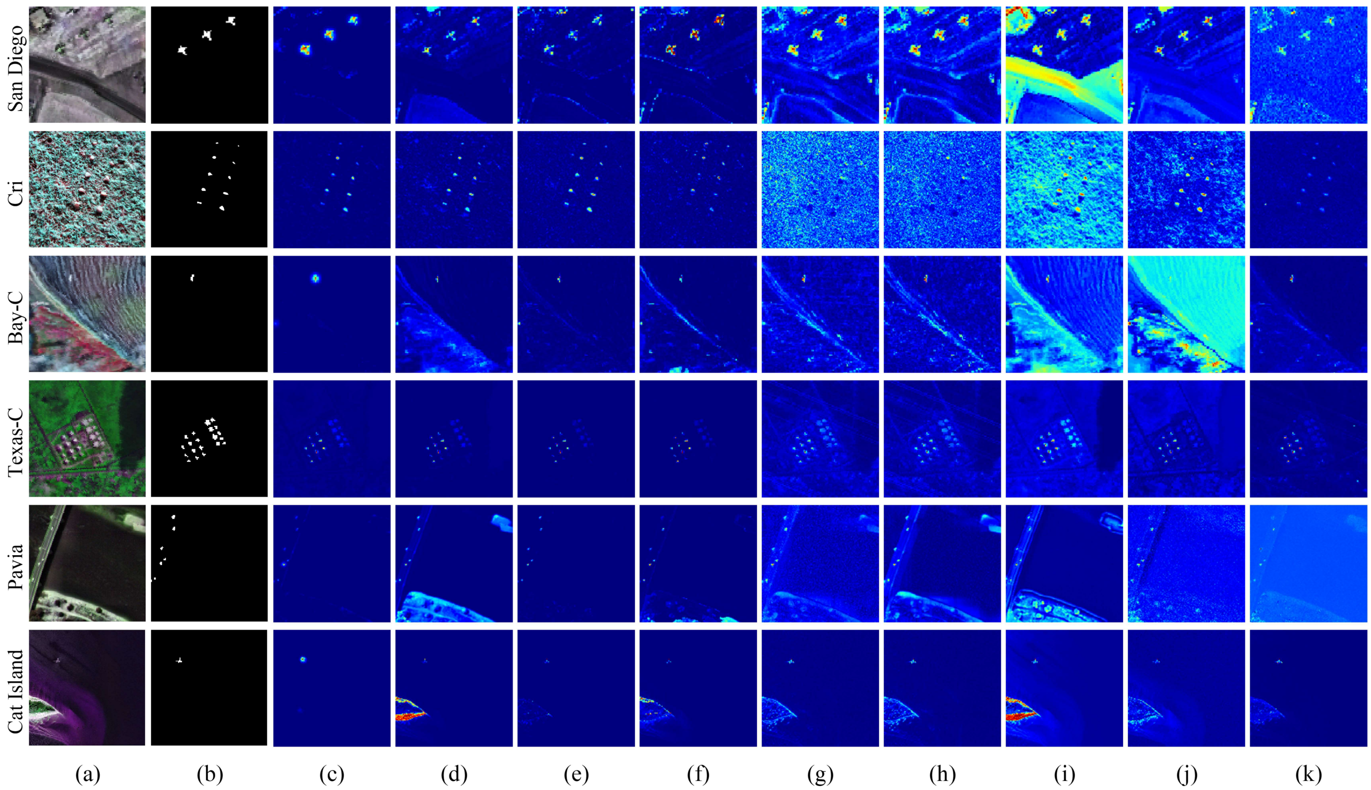

Fig. 7. Qualitative heat maps of (a) false-color image and (b) ground truth. Detection results obtained by (c) proposed SuperB-QSNet, (d) RGAE, (e) PDBSNet, (f) SuperRPCA, (g) GNLTR, (h) PCA-TLRSR, (i) PTA, (j) LRASR, and (k) FrFE-RX.

noting that SuperB-QSNet consistently generates the cleanest detection maps, showcasing well-defined anomaly regions with minimal false alarms. The proposed model demonstrates high alignment with the ground-truth References across all datasets.

Among these HSIs, PDBSNet performs similar to the proposed model, particularly in the Cri and Bay Champagne (Bay-C) datasets, where it effectively isolates anomalies while minimizing background noise. In contrast, other methods show varying performance limitations. For instance, PTA and LRASR generate significant background artifacts and false positives, especially in San Diego and Bay-C scenes, where they mistakenly highlight areas that are not anomalous. On the other hand, the GNLTR and PCA-TLRSR algorithms perform moderately but are affected by noticeable noise interference in complex environments, such as Texas-C. The results from Cat Island further illustrate this issue, as multiple algorithms (RGAE, SuperRPCA, GNLTR, and PCA-TLRSR) incorrectly identify linear features that do not exist in the ground truth. However, both SuperB-QSNet and PDBSNet successfully avoid these artifacts.

2) ROC Curves: In Fig. 8, each subfigure displays the ROC curves obtained by distinct HAD models on the same dataset. Among all the ROC curves, SuperB-QSNet consistently demonstrates superior detection capabilities. It maintains steady curves with smoother transitions than competitors, indicating robust detection stability across varying thresholds, particularly in critical low false alarm regions. Specifically, the corresponding curve rises sharply near the origin in San Diego, outperforming other methods in early detection capability. While several algorithms achieve near-perfect detection on Cri, SuperB-QSNet maintains its advantage through more consistent curve progression. Besides, the matching curve exhibits remarkable separation from most competitors on the Bay-C dataset, particularly from algorithms like LRASR and RGAE that hold significantly flatter trajectories. Furthermore, our model demonstrates an exceptional ability to maintain high detection probability even at extremely restrictive false alarm thresholds on Texas-C. Similarly, its curve consistently dominates the upper left quadrant of the ROC space for both Pavia and Cat Island.

3) AUC, Running Time, and MR: Table I displays various algorithms' AUC and running time performance on six HSIs. The comprehensive evaluation demonstrates that SuperB-QSNet achieves an exceptional balance of accuracy and efficiency across all datasets. This proposed model consistently maintains high accuracy, with AUC scores exceeding 0.9900, while completing detection tasks within 10.00 s. No other algorithm can match this combination of speed and precision. As illustrated in Table I, PDBSNet achieves similar accuracy as SuperB-QSNet on some datasets but requires much longer processing time (up to 1349 s on Cri). RGAE delivers good results on some datasets but performs poorly on others, while consuming extremely long processing time. Traditional approaches like FrFE-RX occasionally work well but lack consistency in performance and efficiency. Considering the comprehensive evaluation of AUC results and running time, our SuperB-QSNet effectively balances detection accuracy

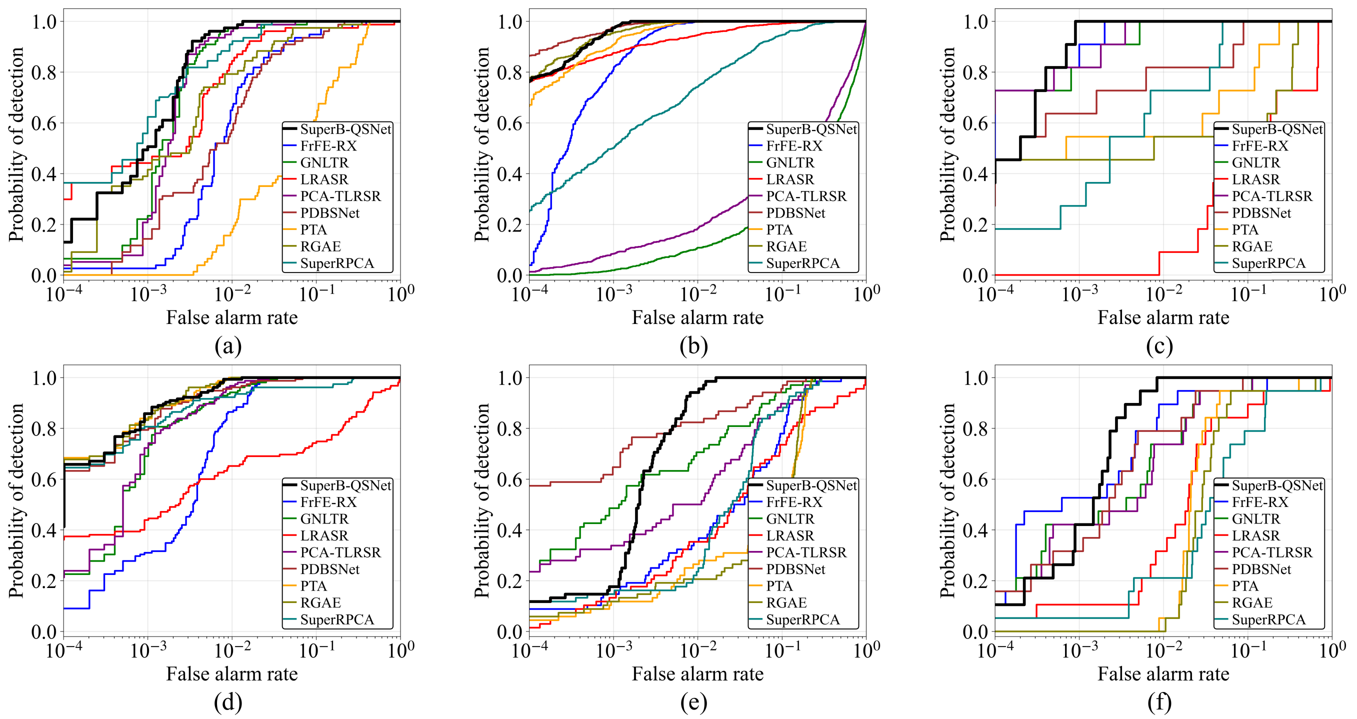

Fig. 8. Qualitative ROC curves by different HAD models on (a) San Diego, (b) Cri, (c) Bay-C, (d) Texas-C, (e) Pavia, and (f) Cat Island.

TABLE I

AUC AND RUNNING TIME (SECONDS). THE BEST AND SECOND VALUES ON EACH DATASET ARE MARKED IN RED AND BLUE, RESPECTIVELY

Methods	San Diego		Cri		Bay-C		Texas-C		Pavia		Cat Island	
	AUC	Time	AUC	Time	AUC	Time	AUC	Time	AUC	Time	AUC	Time
SuperB-QSNet	0.9983	3.621	0.9999	9.372	0.9998	3.152	0.9993	3.111	0.9969	3.523	0.9981	2.824
RGAE	0.9860	147.4	0.9998	1005	0.8668	175.2	0.9994	164.4	0.9042	123.9	0.9385	418.5
PDBSNet	0.9868	82.19	0.9999	1349	0.9850	184.0	0.9982	130.0	0.9883	83.08	0.9906	99.60
Super-RPCA	0.9969	9.103	0.9805	11.24	0.9863	13.99	0.9902	15.32	0.9555	15.85	0.9106	36.07
GNLTR	0.9967	3.543	0.6109	84.81	0.9993	4.264	0.9979	4.284	0.9795	10.69	0.9880	10.53
PCA-TLRSR	0.9970	1.920	0.6939	142.2	0.9995	6.160	0.9983	14.20	0.9644	6.983	0.9875	81.07
PTA	0.8892	26.91	0.9997	199.7	0.9484	36.67	0.9993	44.22	0.9061	64.82	0.9569	110.3
LRASR	0.9804	48.03	0.9960	313.8	0.7580	63.40	0.8876	97.88	0.8777	198.8	0.9239	297.0
FrFE-RX	0.9780	13.03	0.9993	87.60	0.9997	28.45	0.9951	20.63	0.9411	19.27	0.9883	84.66

TABLE II

MRS' COMPARISON BETWEEN DEEP LEARNING ALGORITHMS

Dataset	**SuperB-QSNet**	**PDBSNet**
San Diego	175.72 MB	270.61 MB
Cri	1641.0 MB	615.32 MB
Bay-C	358.29 MB	519.43 MB
Texas-C	436.91 MB	542.66 MB
Pavia	499.63 MB	729.99 MB
Cat Island	162.13 MB	751.16 MB

and computational efficiency, making it a practical choice for real-world HAD applications.

To comprehensively evaluate the proposed algorithms' practical applicability and deployment feasibility, we analyzed memory usage using a sophisticated monitoring framework that employs Python's psutil library. As shown in Table II, MRs vary significantly across different datasets, ranging from 162 to 1641 MB. PDBSNet demonstrates superior memory efficiency on high-dimensional datasets, particularly Cri, where it achieves 62.5% memory reduction compared to SuperB-QSNet. Conversely, SuperB-QSNet exhibits advantages on most datasets, requiring substantially less memory on Cat Island. These results demonstrate that both algorithms are practically deployable on modern computing systems with standard RAM configurations (2–4 GB), providing essential guidance for practitioners in resource planning and system design for real-world HAD applications. Nevertheless, considering other evaluation metrics, our proposed SuperB-QSNet outperforms the alternatives.

4) Separability Box Plots: Fig. 9 presents box plots with outliers in red circles of various anomaly detection methods across six HSIs. The proposed SuperB-QSNet exhibits outstanding performance, consistently maintaining low background scores while achieving effective anomaly

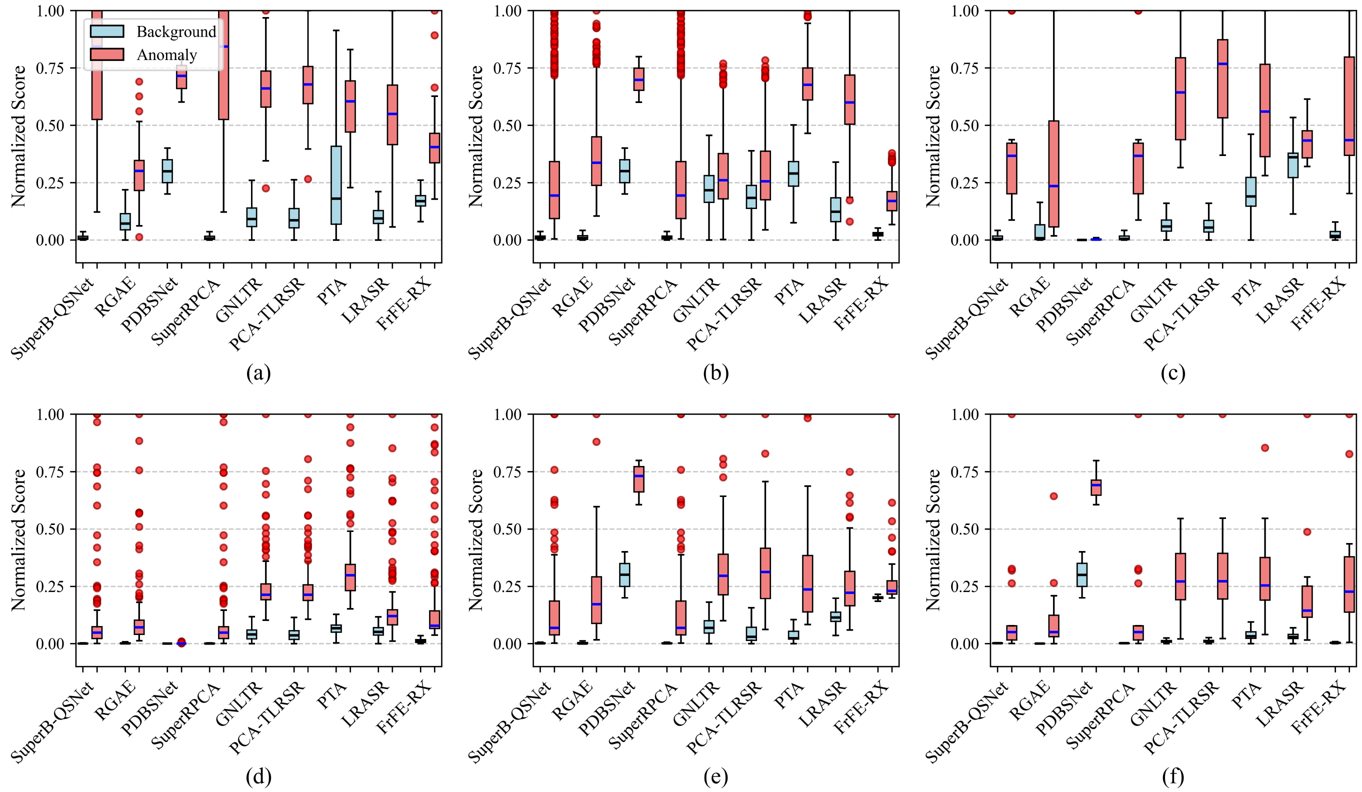

Fig. 9. Separability box plots with outliers in red circles generated by various HAD models on (a) San Diego, (b) Cri, (c) Bay-C, (d) Texas-C, (e) Pavia, and (f) Cat Island.

separation, particularly in San Diego, Bay-C, and Texas-C datasets. The red circle outliers in the anomaly boxes reveal the core anomalous pixels that receive extremely high scores, demonstrating strong discriminative capability despite the compressed appearance of the interquartile ranges caused by the inclusion of lower scoring peripheral pixels within the ground-truth masks. SuperRPCA demonstrates comparable patterns of separation, underscoring the effectiveness of superpixel-based approaches. Within the deep learning category, PDBSNet stands out in Pavia and Cat Island due to its remarkable anomaly-background separation. In contrast, RGAE performs well in Cri but lacks consistency across other datasets. Tensor-based methods, such as GNLTR and PCA-TLRSR, perform better in Bay-C than anticipated, with high anomaly scores that rival those of leading methods.

D. Robustness to Additional Noise

In the model construction, we assume that the acquired images are free from sparse noise, an assumption that often does not hold in real-world raw data. To evaluate the robustness of SuperB-QSNet against noise interference, we introduce two common types of noise with two degrees of intensity to the original HSIs. The first type is independent and identically distributed (i.i.d.) Gaussian noise, denoted as $N(0, \sigma^2)$, with standard deviations $\sigma = 0.05$ and $\sigma = 0.07$. The second type is non-i.i.d. Gaussian noise, where the SNR follows a uniform distribution between [10, 20] dB or [20, 30] dB for each spectral band. These noise types and intensity levels are selected by practical considerations in hyperspectral imaging. The i.i.d. Gaussian noise represents moderate-to-high sensor noise commonly encountered in real-world acquisitions. The non-i.i.d. Gaussian noise with varying SNR levels simulates more realistic scenarios where different spectral bands experience varying noise levels due to atmospheric absorption, sensor characteristics, and illumination conditions. The simulation experiments are conducted using San Diego dataset, where we compare SuperB-QSNet with two existing deep learning approaches: RGAE [40] and PDBSNet [21]. In this experiment, to fairly compare the anomaly detection performance of different algorithms on the same dataset with additional Gaussian noise, we employ the same quantile threshold (99%) to the anomaly score maps produced by each method. Specifically, we mark the top 1% pixels with the highest anomaly scores as anomalous pixels for each technique. This approach ensures that the number of detected anomalies is consistent across methods, allowing for a direct comparison of the distribution of true and false anomaly detections under noise. Consequently, it provides a clear basis for evaluating the robustness of different algorithms to noise perturbations.

The results of the noise robustness evaluation are presented in Fig. 10. SuperB-QSNet exhibits superior detection accuracy, as its predicted target area aligns more closely with the target across various noise conditions. In contrast to PDBSNet, which generates fewer false detections,

TABLE III
AUC BY DIFFERENT MODULES OF SUPERB-QSNET

HH-SBC	QSR	DG-QCD	San Diego	Cri	Bay-C	Texas-C	Pavia	Cat Island
✓			0.9796	0.9991	0.9990	0.9962	0.9597	0.9586
	✓		0.9676	0.9980	0.7226	0.9945	0.9629	0.4735
		✓	0.5532	0.9980	0.9477	0.9990	0.8741	0.9544
✓	✓		0.9688	0.9979	0.9909	0.9949	0.9618	0.5263
✓		✓	0.5750	0.9997	0.9552	0.9992	0.8845	0.9569
	✓	✓	0.9732	0.9994	0.9995	0.9990	0.9947	0.9900
✓	✓	✓	**0.9983**	**0.9999**	**0.9998**	**0.9993**	**0.9969**	**0.9983**

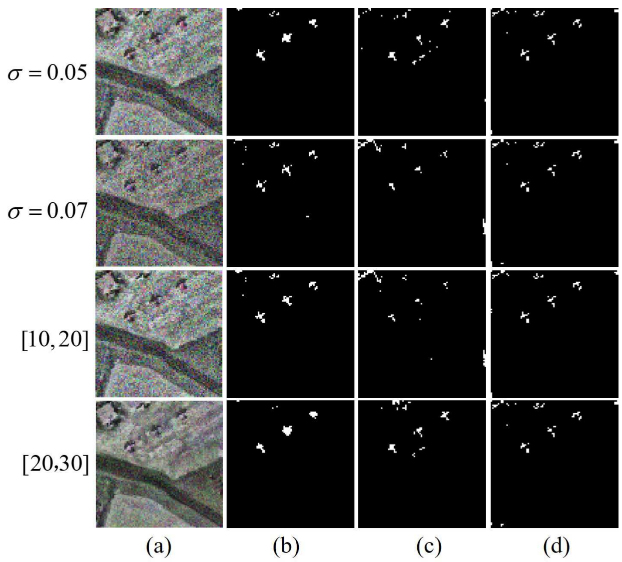

Fig. 10. (a) False-color image of simulated noisy San Diego. Binary maps acquired by (b) SuperB-QSNet, (c) RGAE, and (d) PDBSNet.

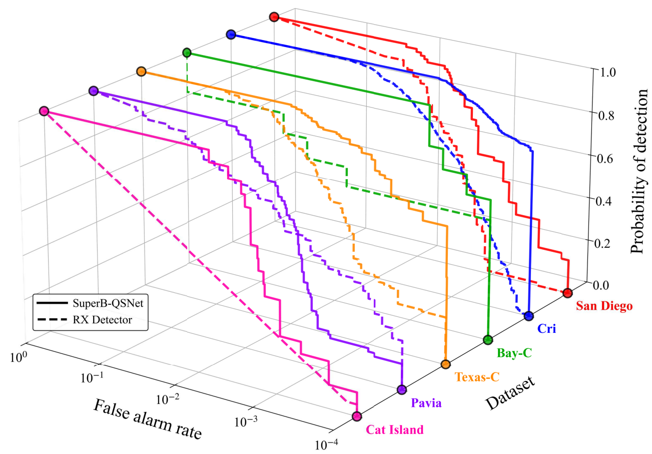

Fig. 11. 3D-ROC on six HSIs acquired by SuperB-QSNet and direct RX detection on sparse components generated after HH-SBC and QSR.

SuperB-QSNet maintains a more balanced performance, achieving comparably low false positives and higher precision. SuperB-QSNet demonstrates strong detection capabilities even in high-noise environments, highlighting its algorithmic adaptability. Unlike RGAE, SuperB-QSNet offers greater stability in noisy conditions and results in significantly fewer errors, further validating the effectiveness of its design. The visual detection results show that our model effectively identifies anomalous targets, even in noisy conditions.

E. Empirical Analysis

1) Ablation Study: To validate the contribution of each component in our proposed model, we conducted a comprehensive ablation study using two different detectors: the DG-QCD and RX detectors. The experiments systematically evaluate the effectiveness of each module using both the classical RX detector and our proposed DG-QCD detector. We chose the RX detector as our baseline because it is a classical statistical anomaly detector that relies solely on second-order statistics (covariance matrix) [11]. This ensures that any performance improvements observed when incorporating HH-SBC or QSR can be attributed exclusively to the algorithmic contributions of these modules, rather than to any effects from neural network optimization. Table III presents the ablation results, where the classical RX detector is the default detection head when DG-QCD is not employed (i.e., when the DG-QCD column is unchecked).

As shown in Table III, the combination of HH-SBC with the RX detector achieves robust performance (AUC >0.95 on most datasets), demonstrating effective spectral redundancy reduction while preserving relevant anomaly information. QSR exhibits dataset-dependent performance; it excels on some datasets but struggles on others with complex spectral characteristics. Fortunately, combining HH-SBC and QSR mitigates QSR's instability issues, indicating that hierarchical band clustering offers a more suitable input structure for sparse decomposition. When DG-QCD is introduced, all combinations show significant improvements, with the complete model achieving optimal performance across all datasets.

Furthermore, a 3D-ROC visualization is created to verify the effectiveness of our proposed method by applying the RX detector and DG-QCD to the sparse components generated by HH-SBC and QSR, as shown in Fig. 11. The 3D-ROC visualization offers several advantages over traditional 2D-ROC curves for multidataset performance evaluation. It enables simultaneous detection performance comparison across multiple datasets within a comprehensive view, facilitating direct visual assessment of method consistency and robustness. Across all six datasets, SuperB-QSNet (solid lines) consistently outperforms the RX detector applied

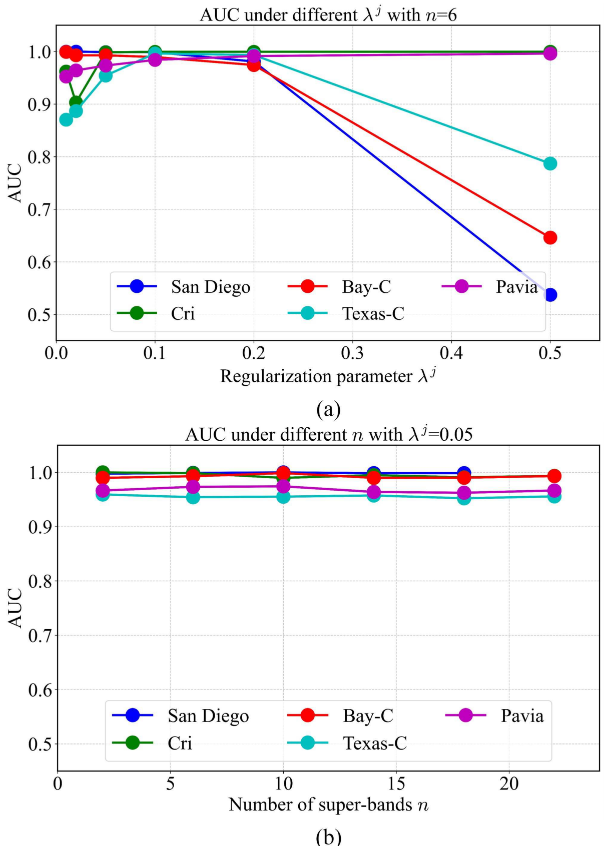

Fig. 12. Parameter sensitivity analysis of SuperB-QSNet on (a) regularization parameter λ^j and (b) number of super-bands n.

to sparse components (dashed lines), demonstrating that SuperB-QSNet maintains consistent performance advantages across all datasets, indicating strong generalization capabilities and robustness. The 3-D visualization also shows that SuperB-QSNet achieves higher true positive rates at extremely low FARs (10^{-4}). This finding is particularly significant for HAD, where minimizing false alarms is crucial for operational deployment.

2) Hyperparameter Sensitivity: To analysis the effect of different parameter values, we tested the performance of the model with the regularization parameter value λ^j ranging from $\{0.01, 0.02, 0.05, 0.1, 0.2, 0.5\}$ and the number of super-bands n ranging from $\{2, 6, 10, 14, 18, 22\}$ on most datasets (except Cri where the maximum of n can only reach 18 because of its limited spectral size). The parameter sensitivity analysis is given in Fig. 12.

Fig. 12(a) illustrates the impact of different regularization parameters on model performance under a fixed super-band number $n = 6$. When the regularization parameter is small, the AUC values across all datasets remain above 0.95, demonstrating excellent stability. However, as the parameter increases to 0.5, significant performance degradation is observed on San Diego, Bay-C, and Texas-C datasets. In contrast, the Pavia and Cri datasets maintain relatively stable performance. This suggests that the choice of regularization parameter has a more pronounced effect on complex scenarios, with the optimal parameter range between 0.05 and 0.1. Fig. 12(b) reveals that with an appropriately set regularization parameter $\lambda^j = 0.05$, the model exhibits minimal sensitivity to variations in super-band numbers, maintaining stable AUC values above 0.95 across all datasets. This robustness confirms that the SuperB-QSNet algorithm possesses strong generalization capabilities and parameter insensitivity, enabling it to adapt to diverse hyperspectral data types without requiring meticulous parameter tuning. Notably, even under the least favorable parameter combinations, the model consistently delivers high performance on Pavia and Cri datasets, further attesting to its reliability.

V. Conclusion

The proposed SuperB-QSNet introduces an innovative framework that synergistically combines spectral-structured processing with quaternion-based techniques for HAD. Our approach addresses the critical limitation of spectral feature underutilization in existing methods through three key innovations: 1) HH-SBC that intelligently groups spectral bands while preserving discriminative information; 2) QSR model that uniquely encodes both spatial and spectral correlations in a unified hypercomplex domain; and 3) DG-QCD that enhances discriminative power through spectral coherence preservation and intergroup difference operations. Through extensive experimentation, we demonstrate that this integrated approach significantly improves the identification of subtle spectral anomalies while maintaining computational efficiency, even under noisy scenes. In the future, we will focus on developing dynamic or irregular size-adapted super-unit quaternion operations that automatically balance spectral and spatial feature representation without manual tuning. Moreover, we plan to expand the manner of identifying and maintaining spectral context to the domain of multitemporal or multimodal analyses.

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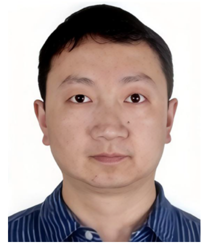

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